

Compounds of Transitional elements

161. Which of the following sulphides is yellow in colour
 (a) CuS (b) CdS
 (c) ZnS (d) CoS
162. Which of the following is not oxidized by O_3
 (a) KI (b) $FeSO_4$
 (c) $KMnO_4$ (d) K_2MnO_4
163. The number of moles of $KMnO_4$ reduced by one mole of KI in alkaline medium is
 (a) One fifth (b) Five
 (c) One (d) Two
164. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is **incorrect** for this reaction
 (a) $Na_2S_2O_3$ is oxidised
 (b) CuI_2 is formed
 (c) Cu_2I_2 is formed
 (d) Evolved I_2 is reduced
165. The only cations present in a slightly acidic solution are Fe^{3+} , Zn^{2+} and Cu^{2+} . The reagent that when added in excess to this solution would identify and separate Fe^{3+} in one step is
 (a) $2M HCl$ (b) $6M NH_3$
 (c) $6M NaOH$ (d) H_2S gas
166. Which element is alloyed with copper to form bronze
 (a) Fe (b) Mn
 (c) Sn (d) Zn
167. Emery consists of
 (a) Impure corundum (b) Impure carborundum
 (c) Impure graphite (d) Purest form of iron
168. The metal commonly present in brass and german silver is
 (a) Mg (b) Zn
 (c) C (d) Al
169. In the equation
 $4M + 8CN^- + 2H_2O + O_2 \longrightarrow 4[M(CN_2)]^- + 4OH^-$ The metal M is
 (a) Copper (b) Iron
 (c) Gold (d) Zinc
170. The term plating is
 (a) Platinum painting
 (b) Flat sheet of platinum
 (c) Platinum manufacturing
 (d) Platinum used as a catalyst
171. Purple of cassius is
 (a) Gold solution
 (b) Silver solution
 (c) Copper solution
 (d) Platinum solution
172. Match the items under **List 1** with the compounds/elements from the



List 2. Select the correct answer from the sets (a), (b), (c) and (d).

List 1

List 2

- (i) Explosive
(ii) Artificial gem
(iii) Self reduction
(iv) Magnetic material

- (A) NaN_3
(B) Fe_3O_4
(C) Sn
(D) Al_2O_3
(E) $Pb(N_3)_2$
(F) Fe_2O_3
(G) Cu
(H) SiC

- (a) (i) A, (ii) D, (iii) G, (iv) B (b) (i) A, (ii) D, (iii) G, (iv) F
(c) (i) E, (ii) D, (iii) G, (iv) B (d) (i) E, (ii) H, (iii) C, (iv) F

173. Blood haemoglobin contains the metal
(a) Al (b) Mg
(c) Cu (d) Fe

174. Percentage of carbon in steel is
(a) 2.5 – 4.5%
(b) 0.25 – 0.5%
(c) 0.2 – 1.5% (d) 3 – 3.5%

175. Steel is manufactured from
(a) Wrought iron (b) Cast iron
(c) (a) and (b) both (d) Haematite

176. Modern method for the manufacture of steel is
(a) Bessemer process
(b) Seimen-Martin's open hearth process
(c) Duplex method

(d) L.D. process

177. **Spiegeleisen** is an alloy of

- (a) Fe , C and Mn
(b) Fe , Mg and C
(c) Fe , Co and Cr
(d) Fe , Cu and Ni

178. Stainless steel is an alloy steel of the following metals

- (a) Fe Only (b) Cr and Ni
(c) W and Cr (d) Ni and Be

179. In the manufacture of steel, the Bessemer converter is containing lining of

- (a) SiO_2 (b) CaO
(c) CaO and MgO (d) Fe_2O_3

180. Which of the following alloys contain only Cu and Zn

- (a) Bronze (b) Brass
(c) Gun metal (d) Bell metal

